HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and integrated **electronic circuits** and subsystems for active implantable medical devices, contributing to hardware integration with embedded software.
- Assisted in **schematic capture**, circuit simulation, and **PCB development**, focusing on component selection under senior engineer guidance.
- Participated in electrical characterisation and performance evaluation of hardware components using standard **laboratory test equipment**.
- Collaborated with **Systems, Software, and Mechanical** engineering teams to support product development lifecycle and technical problem-solving.
- Generated and maintained **engineering documentation** in line with Quality Management System requirements, ensuring traceability and compliance.
- Supported **risk management activities** and the implementation of design mitigations, working closely with the project team to address potential issues.
- Containerized backend services using **Docker** and deployed to **AWS EC2** instances, streamlining the deployment process for internal tools.

SKILLS

Hardware Engineering – Electronic Engineering, Electrical Engineering, Schematic Capture, Circuit Simulation, PCB Development, Component Selection, Electrical Characterisation, Hardware Integration

Software & Systems – Embedded Software, Systems Engineering, Quality Management System, Risk Management, Design Controls

Programming & Scripting – Python, Java, SQL, Shell Scripting

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Change Management, Linux

PROJECTS

Medical Device Electronic Subsystem Development

- Designed and simulated **electronic circuits** for a novel implantable device, focusing on low-power consumption and signal integrity.
- Assisted in the **PCB layout** process and component selection, ensuring compliance with medical device standards and material requirements.
- Conducted initial **electrical characterisation** tests on prototype subsystems, documenting performance metrics and identifying areas for refinement.

Automated Test System for Embedded Hardware

- Developed a **Python**-based automation framework to control test equipment and execute verification routines for electronic components.
- Integrated the testing script with **Jenkins** for automated build and test cycles, reducing manual testing effort by approximately 40%.
- Captured and analysed test results, generating reports on component performance against specifications using **pandas**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)